



Masaya Volcano

A huge volcano in Nicaragua that sometimes features a rare, spectacular lake of fiery lava

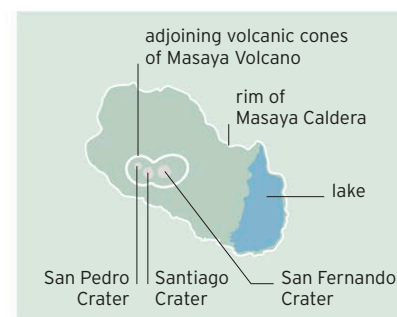
Masaya is an example of a complex volcano: it has more than one volcanic cone, as well as multiple vents and craters. Its two main cones lie within a larger, roughly oval structure called the Masaya Caldera, which formed during a cataclysmic eruption some 2,500 years ago. The caldera is about 7 miles (11km) long, 3 miles (5km) wide, and has a rim that in places is 980 ft (300m) high.

At the summit of Masaya's main cones are several deep holes called pit craters. One of these, the Santiago Crater, has been the main site of volcanic activity at Masaya for many years and sometimes contains a red-hot pool known as a lava lake.



THE MASAYA CALDERA

Masaya Volcano lies within the much larger Masaya Caldera. The volcano itself consists of two main overlapping volcanic cones. One contains at its summit two pit craters—the Santiago and San Pedro craters—while the other is indented by the larger San Fernando Crater.

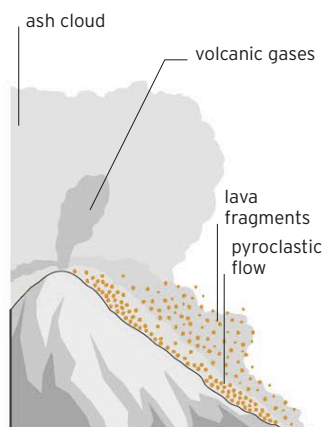


△ HELLISH PIT

The Santiago Crater sometimes fills with a lava lake that produces spectacular bursting bubbles as it boils off noxious gases. It is one of just a handful of lava lakes in the world.

PYROCLASTIC FLOWS

Also known as a *nuée ardente* (glowing cloud), a pyroclastic flow is an avalanche of volcanic gases, ash, and lava fragments. It often forms from the collapse of a volcanic eruption cloud or a lava dome. Most pyroclastic flows travel for about 3–6 miles (5–10 km), typically moving downhill at around 60 mph (100 kph) or faster, burning, flattening, and burying everything they encounter along the way.



Soufrière Hills

A Caribbean volcano that has devastated over half the island of Montserrat since 1995



There are several volcanoes named Soufrière in the Caribbean—*soufrière* is French for “sulfur outlet”—but the most famous is on the island of Montserrat, called Soufrière Hills. Since 1995, several destructive events called pyroclastic flows (see panel, left) have caused it to be the most studied volcano in the world.

Collapsing lava domes

Soufrière Hills produces a viscous lava that, instead of flowing away, piles up to form a lava dome. This emits a lot of steam and sometimes glows at night. The danger with lava domes, though, is that they can disintegrate to produce pyroclastic flows. At Soufrière Hills, a series of dome collapses and flows between 1995 and 2000 destroyed Montserrat's capital city, Plymouth, obliterated many settlements, and caused several deaths. Two-thirds of the population had to leave the island. However, since 2010, the volcano has quieted, and Montserrat remains a compelling place to visit because of its scenic beauty and volcanic history.



△ DANGEROUS DOME

The gray conical object in the background here is a lava dome. Part of its flank has begun to break up, triggering a pyroclastic flow, which resembles an avalanche.